

Troubleshooting Methodology -- Guided Notes ANSWER KEY

Unit tp-1.8 | CompTIA Network+ N10-009 Obj. FC0-U71 1.4 | N10-008 FC0-U71 1.4

For instructor use / distribute after students complete the notes.

Question 1

The CompTIA troubleshooting methodology has _____ steps. Step 1 is _____. Step 2 is _____. Step 3 is _____. Skipping steps -- especially jumping from Step 1 directly to implementing a fix -- often results in _____ the wrong problem.

Answer: 7; identify the problem; establish a theory of probable cause; test the theory; solving (fixing).

Question 2

In Step 1 (identify the problem), a technician should ask _____ -ended questions (e.g., 'What happens when you turn it on?') rather than _____ questions (e.g., 'Does it turn on?'). Open questions gather _____ detail and lead to better theories in Step 2.

Answer: Open; closed; more.

Question 3

Step 4 is to establish a/an _____ of action to resolve the problem, including identifying potential _____ of the planned fix. This prevents a technician from solving one problem while creating another. Step 5 is to _____ the solution and then _____ full system functionality.

Answer: Plan; effects (side effects); implement; verify.

Question 4

Step 6 is to _____ findings, actions, and outcomes. This creates a reference for future similar issues and protects the technician legally if decisions are later questioned. Step 7 is to _____ the user -- explaining what caused the problem and how to prevent it in plain language.

Answer: Document; educate.

Question 5

The distinction between a _____ (what the user reports, e.g., 'the screen is black') and a _____ (the actual technical cause, e.g., 'the GPU is unseated') is critical. Fixing a symptom without identifying the _____ will result in the problem _____.

Answer: Symptom; root cause; root cause; recurring (returning).

Question 6

When testing a theory in Step 3 and the theory is _____, the technician should either establish a new theory or _____ to a higher skill level. Before escalating, the technician should document _____ steps already tried so the next technician does not repeat them.

Answer: Disproved (not confirmed); escalate; all.

Question 7

When questioning a user in Step 1, a technician should ask whether any _____ were made recently -- such as software updates, new hardware, or changed settings. These changes are often the _____ of sudden failures. The technician should also reproduce the problem if possible to _____.

Answer: Changes; cause (trigger); confirm the symptom (verify it themselves).

Question 8 -- Real World Application

REAL WORLD: A technician skips steps and immediately reinstalls the OS on a laptop with a reported display problem, which takes 3 hours and does not fix the issue. The actual cause was a loose display cable. Identify which troubleshooting steps were skipped, explain what the correct process would have been, and describe the cost of skipping the methodology.

Answer: Skipped steps: 2 (establish theory -- no investigation of probable hardware causes), 3 (test the theory -- no attempt to test with external monitor or check the cable), and 4 (establish a plan -- no consideration of less disruptive fixes first). Correct process: identify symptom (display issue); theory = loose cable or driver issue; test = connect to external monitor. If the external works, the problem is the display panel/cable, not the OS. A 2-minute cable re-seat would have fixed it. Cost of skipping: 3 hours of unnecessary work, loss of user data (if any), user frustration, and the actual problem still unfixed.